SE 1105 - PROGRAMMING AND PROBLEM SOLVING I

Introduction to the Course

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General Remarks

- Theory: Wednesdays from 12:40 to 15:30 in M116 (Section 1)
  Wednesdays from 15:40 to 18:30 in M116 (Section 2)
- Lab: Fridays from 8:40 to 10:30 in Y314 (Section 1)
  Fridays from 10:40 to 12:30 in Y314 (Section 2)
- Office hours: Thursdays from 13:30 to 15:30 (or by email appointments) at M519
- e-mail: kazim.erdogdu@yasar.edu.tr
- moodle: https://online.yasar.edu.tr
- Assessment:
  - Lab: 10%
  - Midterm: 40%
  - Final: 50%
- Programming Language: C

Textbooks

- Problem Solving and Programming Concepts, 9th ed., M. Sprankle, J. Hubbard, Prentice Hall, 2011
- The C Programming Language, B.W. Kernighan, D.M. Ritchie, Englewood Cliffs, Prentice Hall, 1988
- C How to Program, 7th ed., P. Deitel, H. Deitel, Prentice Hall, 2013
- Problem Solving and Program Design in C, 7th ed., J.R. Hanly, E.B. Koffman, Pearson, 2013

Goals

- Understand basic concepts of programming
- Understand how computers handle programs while running (compilers, memory management, function calls, etc.)
- Being able to understand existing programs
- Being able to write our programs solving simple computational problems
- Being able to re-use existing code in our programs

Prerequisites

NOTHING

Participation

Participate in the class by

- asking questions
- answering questions
- sharing ideas
- making corrections
- ...

Computer Programming and Problem Solving

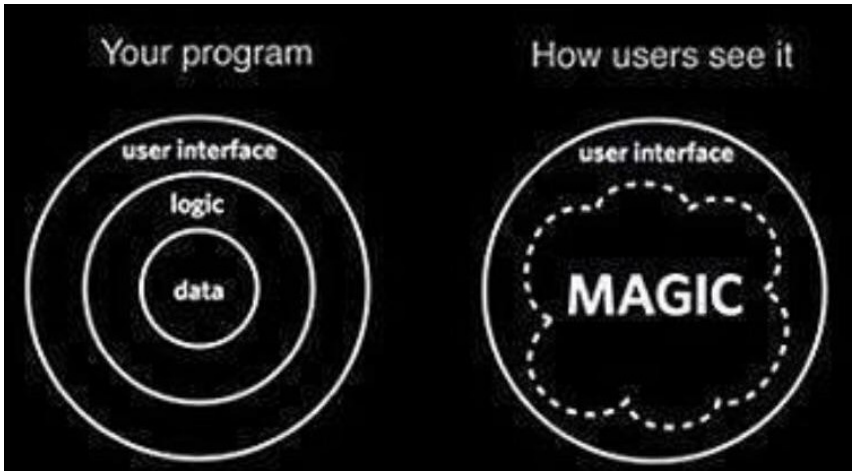


Figure: https://hackernoon.com/30-jokes-only-programmers-will-get-a901e1cea549

Computers are generic devices that can perform any computational task (programs).

Figure: https://www.codementor.io/@fbuireu/35-programming-jokes-that-will-make-you-laugh-1ec7eq11za

Algorithm

"Informally, an algorithm is any well-defined computational procedure that takes some value (or set of values) as input and produces some value (or set of values) as output. An algorithm is thus a sequence of computational steps that transform the inputs to the outputs." (Cormen et al., 2009)

Algorithm: a step-by-step procedure for solving a problem or accomplishing some end, especially by a computer.
(https://www.merriam-webster.com/dictionary/algorithm)

Everyday Examples

- A recipe
- Shopping in a market
- Online shopping
- Construction of a building
- Driving a car
- ...

Mathematical Sequence

5, 18, 17, -6, 7, -8, 58, 0, -4, 16, 43 19, 51, -5, 24, -12, 4

1. What are the minimum and maximum numbers of this sequence?
2. Can you sort them in ascending order?

How did you do them?

Water Jugs Problem

Figure: https://www.reddit.com/r/puzzles/comments/ca9cy6/how_do_you_get_exactly_4_liters_into_the_5_liter/

- You have an infinite water resource and two jugs with 3 and 5 liters capacities.
- Can you get 4 liters of water in 5-liter jug with only the following operations?
  - Empty a jug
  - Fill a jug
  - Pour water from one jug to another until one is either empty or full.

Light Bulbs & Switches

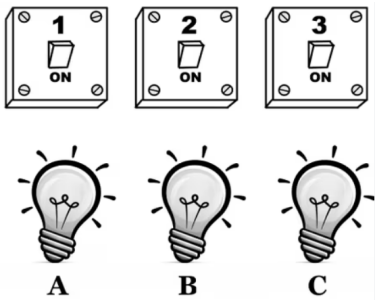


Figure: https://www.theguardian.com/science/2023/aug/07/did-you-solve-it-are-you-smart-enough-for-mensa

- There are 100 lights in a huge room. A switch panel outside the room has 100 switches each corresponding to one light in the room.
- But none of the switches or lights are labeled, so it's difficult to identify which switch turns on which light.
- You are standing outside of the room and your task is to map each switch to its corresponding light using a marker pen.
- You must do this mapping in a minimum number of trips to the room.
- Initially, all switches are turned off.
- What is the minimum number of trips you need to map each switch to its corresponding light?

Source and Solution: https://www.youtube.com/watch?v=ltsx5389iEs&t=38s

Light Bulbs & Switches
Principles

1. Begin with all the lights and switches off.
2. Initially consider all the switches in one group.
3. If switches in a group have the same value:
    1. Divide each group into two even sub groups.
    2. Choose one sub group in each group and change state of all lights and switches in them.
    3. Depending on the current states of all light bulbs and switches, add 0 (for off) or 1 (for on) to their labels.
4. If all lights and switches have unique values, mapping is complete. Stop!
5. Else, choose one of the groups and start a new trip by going to step 3.

Source and Solution: https://www.youtube.com/watch?v=ltsx5389iEs&t=38s

Meet-My-Bot

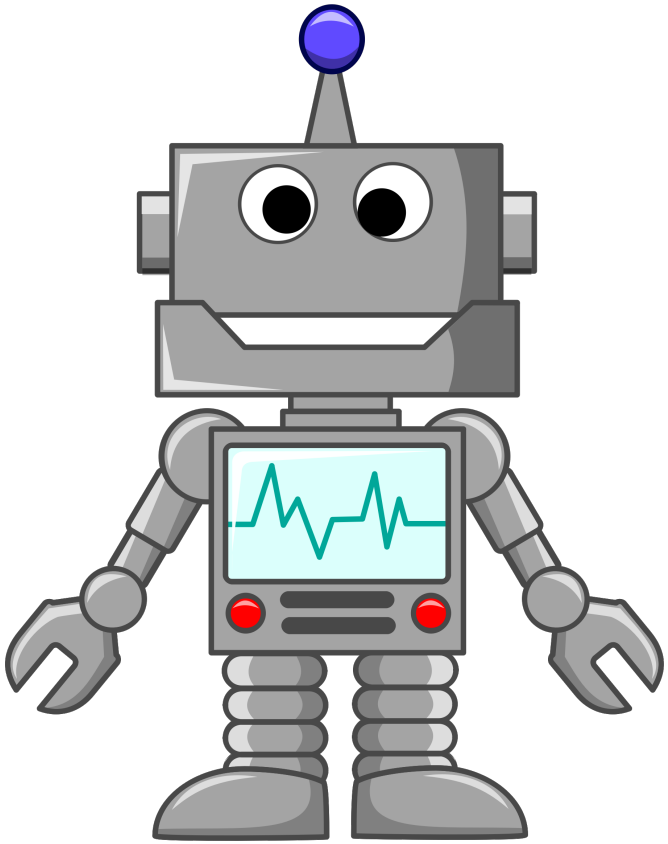


Figure: https://www.clipartmax.com/middle/m2i8i8H7m2G6G6d3_clipart-robot-clipart/

Instructions

000	Wake up
001	Sleep
010	Write on the board
011	Take the pen
100	Drop the pen
101	Raise your arm
111	Lower your arm

How It Works

Figure: https://hpc-wiki.info/hpc/Compiler

How It Works

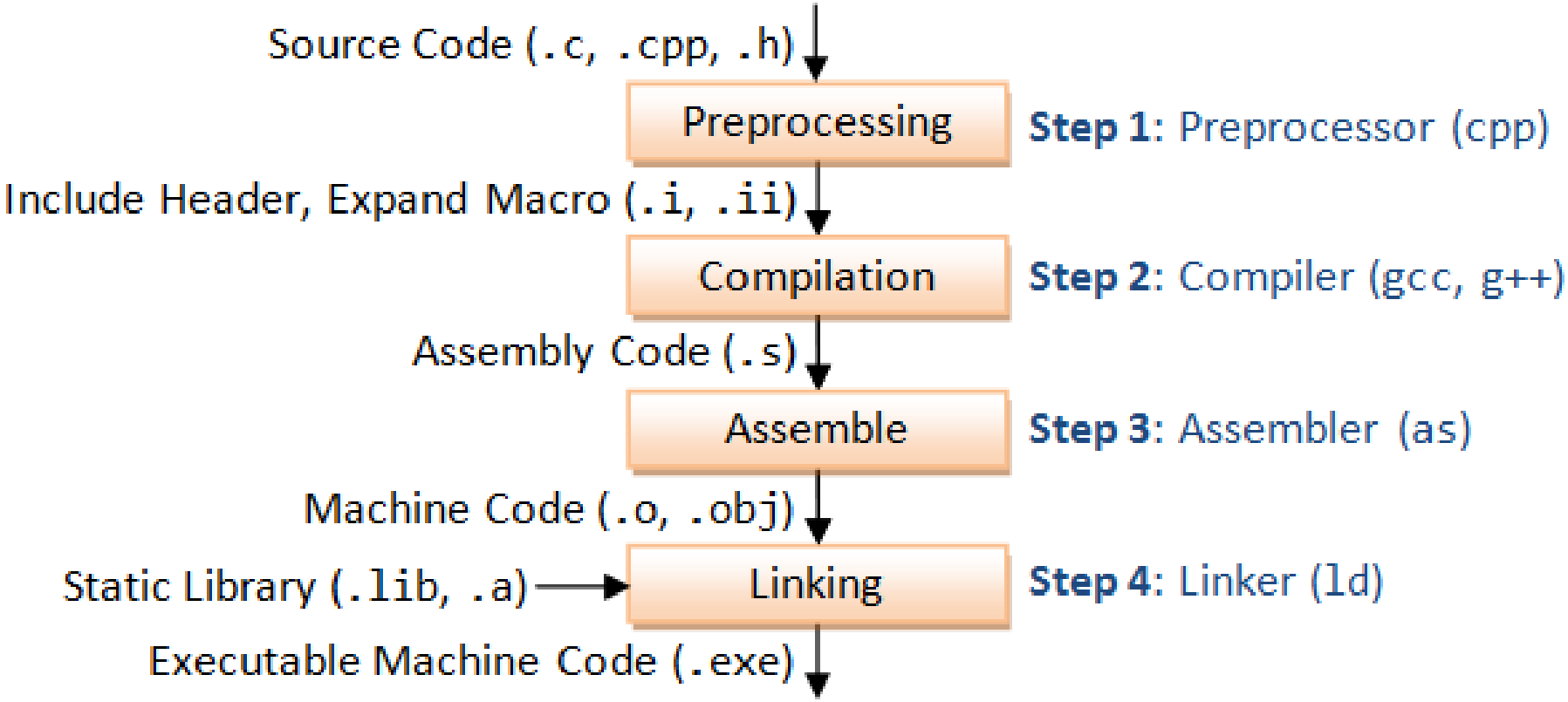
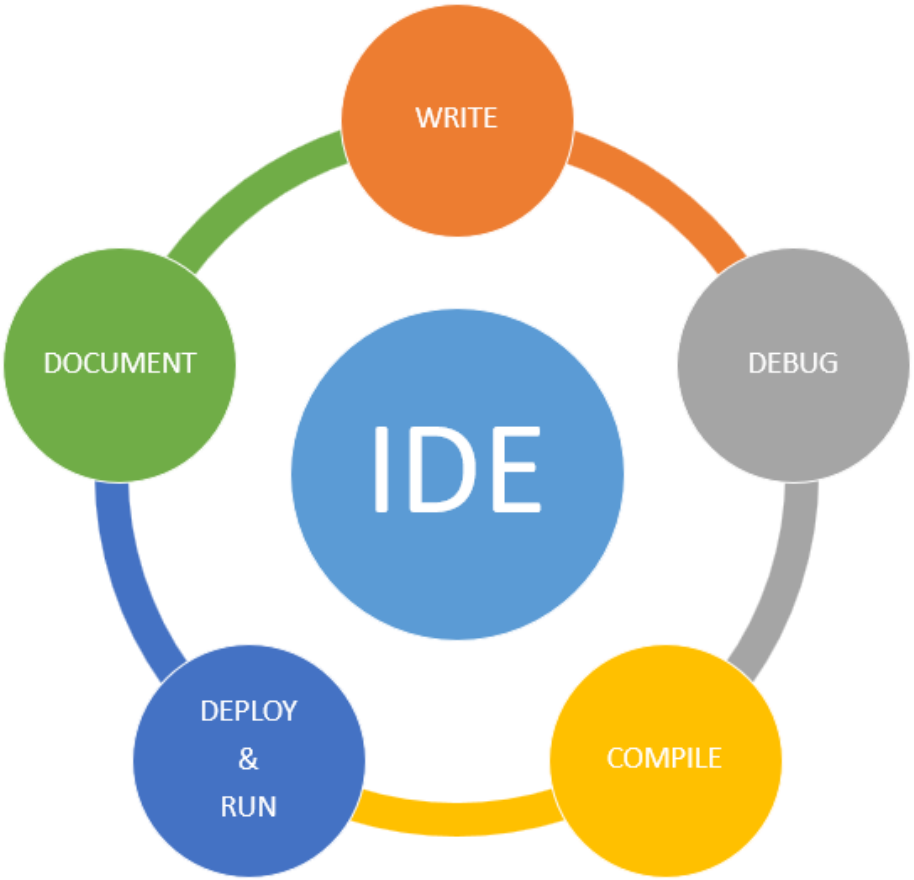


Figure: https://www3.ntu.edu.sg/home/ehchua/programming/cpp/gcc_make.html

Integrated Developement Environment (IDE)



Our First Program: "HelloWorld.c"

HelloWorld.c

```
1  #include <stdio.h>
2  int main(void)
3  {
4      printf("Hello World!");
5      return 0;
6  }
```

References

Cormen, T. H., Leiserson, C. E., Rivest, R. L., and Stein, C. (2009). *Introduction to Algorithms*. MIT Press, 3rd edition.